Form 45 Mar. 2 March 2027 Affinities: rate of increase and changes

1. Warming · Flash $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.

2. Heart · Flash Meaning of variation of $f(x) = -4x + 7$?

3. Heart + · Demonstration Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

4. Back · Table Ages of the students of a club.

Age	10	11	12	13
Count	3	6	10	6

How many students are up to 11 years old? How often?

5. Red thread · Programme → formula Program: multiply by 3 then subtract 2. $f(x)$ then $f(4)$.

Form 46 4 March 2027 Absolute and square value functions: sign and variations

1. Warming · Flash Calculate $|-5|$, $(-5)^2$ and $|2 - 7|$.

2. Heart · Method Give the table of changes in $x \mapsto |x|$ and his sign.

3. Heart + · Reflection Compare $(-3)^2$ and $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?

4. Return · Method We're launching two balanced dice.

5. Red thread · Method Program: choose a number; add 1; raise squared; subtract squared from
What result for 6? What general expression?

Form 47 Mon. 8 March 2027 Reciprocal function: variations; relative position of y = x and y = x^2

1. Warming · Flash Calculate $\dfrac{1}{2}$, $\dfrac{1}{-4}$ and $\dfrac{1}{0{,}1}$.

2. Heart · Method Compare $\dfrac{1}{3}$ and $\dfrac{1}{5}$; what do we say about changes in the reciprocal function on $]0; +\infty[$?

3. Heart + · True/false True or false? Justify. For all real $x > 0$, $x^2 \geqslant x$.

4. Back · Flash What is the front view and the top view of a cylinder placed on its base?

5. Red thread · Prove Show that the product of two integer pairs is a multiple of 4.

Form 48 Mar. 9 March 2027 Compare f (a) *et* $f(b)$; optimization problems

1. Warming · Flash $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

2. Heart · Method $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculation, using the changes.

3. Heart + · Optimization $A(x) = x(10 - x)$ for $x \in [0; 10]$ (area of a perimeter rectangle 20).
that $A(x) = 25 - (x - 5)^2$ and deduce the maximum.

4. Back · Flash 18 out of 24 students have more than 10 averages. Give the frequency in percentage.

5. Red wire · MCQ The solution of $\dfrac{x}{3} + 1 = \dfrac{7}{2}$ is:

A. $\dfrac{15}{2}$ **B.** $\dfrac{9}{2}$ **C.** $\dfrac{21}{2}$ **D.** $\dfrac{5}{6}$

Crossing of two qualitative variables

Form 49 11 March 2027 Crossing of two qualitative variables

1. Warming · Flash Express $\frac{24}{60}$ in percentage.
2. Heart · Flash In a class of 30 students, 18 are girls. Boys?
3. Heart + · Sort Classify each variable: eye colour; baccalaureate mention (AB, B, TB); Size in cm.
4. Back · Problem A rectangle of perimeter 20 cm has for length x. Expressing its area $A(x)$; calculate $A(3)$ and $A(5)$.
5. Red thread · Flash Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.

Form 50 Mon. 15 March 2027 Qualitative variables; cross-table of numbers

1. Warming · Flash Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.
2. Heart · Table Total 40 students. Girls: 22; half boarders (DP): 25; girls DP: 15. Cross table.

	DP	External	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

3. Heart + · True/false True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.
4. Back · Problem Reduction A: -30% out of 80,000 Ar. Reduction B: -20 per cent then -10 per cent. More advantageous?
5. Red thread · Method We know that $AB^2 + AC^2 = BC^2$. What conclusion do we draw? What property?

Form 51 Mar. 16 March 2027 Marginal frequencies

1. Warming · Flash Express $\frac{25}{40}$ in percentage.
2. Heart · Method With the table in form 50: calculate the marginal frequency of girls and girls Half boarders.
3. Heart + · Flash The marginal frequency of girls is 55%. What is that of boys?
4. Back · Flash Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.
5. Red thread · Flash $A(-1;2)$ and $B(3;2)$. What can be said of the right (AB) ? Give AB.

Form 52 18 March 2027 Conditional frequencies

1. Warming · Flash Give $\frac{15}{22}$ in percentage rounded to tenth.
2. Heart · Method With the same picture: among girls, what frequency of DP? Among boys?
3. Heart + · Distinguish Distinguish "the frequency of girls among DPs" and "the frequency of DPs among Girls."
4. Back · Construction programme We're tracking $[AB]$ 6 cm, then center circles A and B and radius 4 cm, which intersect in C and D. Nature of $ACBD$?
5. Red thread · Flash Is 91 first?